

DATA STRUCTURES - 2021

SECTION – A

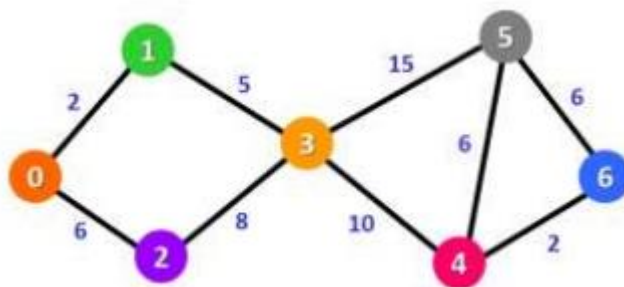
1. How Data Structures are classified?
2. Define ADT and mention its benefits.
3. What are the types of Queue Data Structure?
4. Define Priority Queue.
5. The In-Order Sequence for a Binary Tree is 23,211,89,18,10,20,32. Draw the Binary Tree Structure.
6. What is Trie Structure?
7. Define Bi-Connectivity.
8. Differentiate BFS and DFS.
9. How Quick Sort works?
10. Define Re-hashing.

SECTION – B

11. A. What are the ways to insert a node in the linked list? Write an algorithm to insert a node before a given node in the list.
B. State the difference between arrays and lists.
12. A. Explain the operations on Doubly Linked list in detail.
B. Explain Stack ADT and various operations performed on it using linked list.
13. A. What are the different forms of an expression? Explain the algorithm of Infix to Postfix conversion and also convert $(a + (((b * c) / (d - e))))$ into postfix form.
B. What is B- Tree? List out its properties. Construct B- Tree of order 4 for the list of elements 10, 5, 25, 100, 45, 23, 12, 17, 34, 8, 30, 29, 42, 13.
14. A. Define AVL tree. Explain Single and Double rotations with its suitable examples.
B. Explain in detail about Breadth First Search with suitable example.
15. A. Explain the following terms with an example:
 - i. Cut vertex
 - ii. Euler Circuits
B. Explain Merge Sort with an algorithm. Demonstrate how the following list of elements 10, 100, 23, 45, 67, 89, 12, 15, 78, 64 sorted.

SECTION – C

16. A. Explain Dijkstra's Shortest Path algorithm and find the cost of visiting each node (from vertex 0) for the below graph.



B. What is Heap Sort? Explain the steps involved in sorting 23, 15, 100, 43, 28, 44, 5, 105, 25 using Max-Heap strategy.